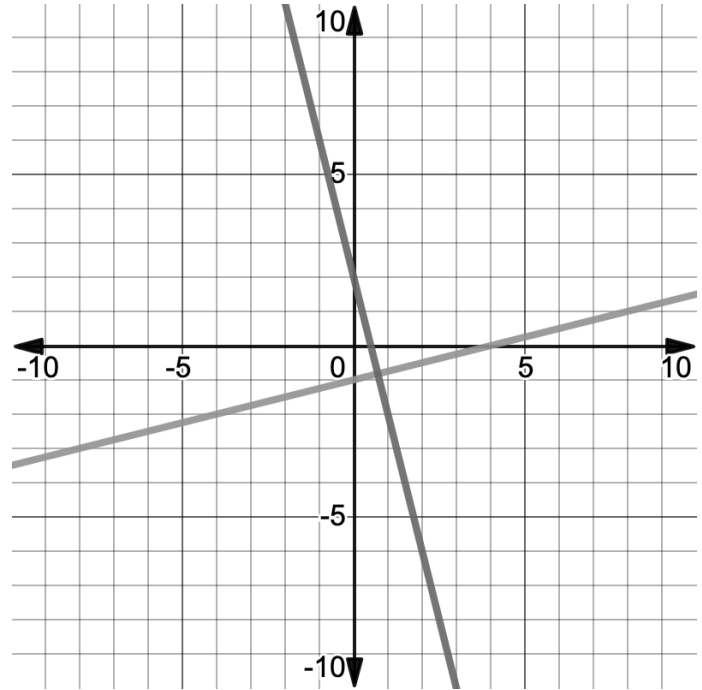


Algebra 1
Unit 4
Lesson 9 Homework

Name **Answer Key** _____
Date _____
Period _____

Use the following information to answer questions 1 through 5.

The lines of the equations $y - 6 = -4(x + 1)$ and $x - 4y = 4$ are shown on the same coordinate grid.



1. What is the slope of the line representing the equation

$$y - 6 = -4(x + 1)?$$

$$y - 6 = -4x - 4$$

$$y = -4x - 4 + 6$$

$$y = -4x + 2$$

$$m = -4$$

2. What is the slope of the line representing the equation $x - 4y = 4$?

$$\frac{x}{4} - \frac{4}{4} = \frac{4y}{4}$$

$$\frac{x}{4} - 1 = y$$

$$m = \frac{1}{4}$$

3. What is the slope of a line that is parallel to $x - 4y = 4$?

$$m = \frac{1}{4}$$

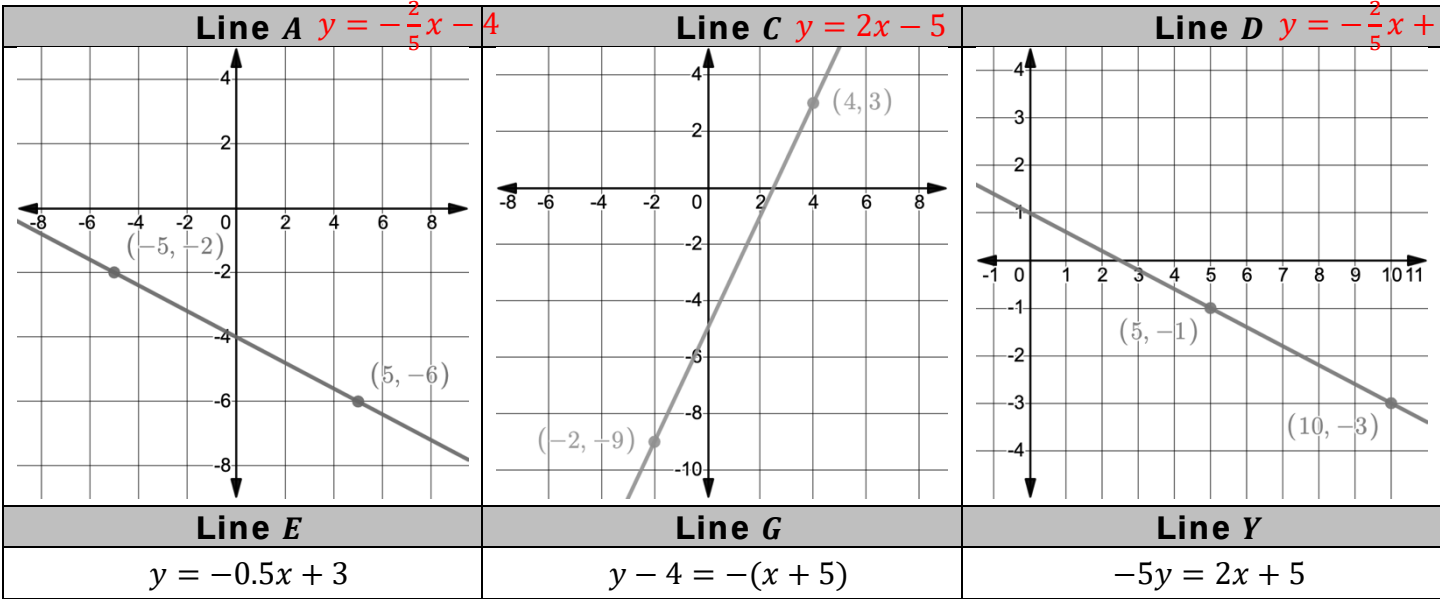
4. Explain how to determine if two lines are parallel.

They have the same slopes but different y -intercepts.

5. Determine if the lines are parallel, perpendicular, or neither.

These lines are perpendicular.

Use the following lines to answer questions 6 through 8.



6. Which lines are parallel?
A, D, & Y
7. Which lines are perpendicular?
C & E
8. What is the slope of a line that is perpendicular to Line G?
1

9. Determine if the lines of the two equations are parallel, perpendicular, or neither.

Equation K	Equation V
$7x - 4y = 8$ $y = \frac{7}{4}x - 2$	$4x - 7y = 14$ $y = \frac{4}{7}x - 2$

Neither

10. Determine if the lines of the two equations are parallel, perpendicular, or neither.

Equation M	Equation R
$3x - 2y = 6$ $y = \frac{3}{2}x - 3$	$y + 8 = \frac{3}{2}(x + 6)$ $y = \frac{3}{2}x + 1$

Parallel